

ELARA: Energy-Efficient and Latency-Aware Service Routing with Adaptive Replica Redeployment for Microservices in Space Computing Power Networks

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Abstract—The rapid advancement of the Space Computing Power Network (SCPN), comprising numerous low Earth orbit (LEO) satellites, has enabled the execution of complex in-orbit observation and computation tasks. To facilitate this, microservice architecture decomposes large, monolithic remote sensing and image processing applications into independent, schedulable, and reusable services deployed in a distributed manner across LEO satellites. However, integrating these capabilities presents significant challenges arising from the limited computing resources, constrained power provisions of individual satellites, and the dynamic nature of space network topology and channel conditions. This paper addresses these challenges by proposing an energy-aware service routing algorithm for microservices within the SCPN. The algorithm aims to minimize the energy consumed by the procedures of processing and transmitting service data in both computation and communication. By modeling the service routing problem as a Directed Steiner Tree (DST) and implementing heuristic-based solutions, our algorithm effectively responds to the diverse service patterns requested by terrestrial clients. Comprehensive experimental results demonstrate that our approach significantly outperforms existing methods in terms of energy efficiency, robustness, stability, and adaptability.

Index Terms—Energy, Microservice, Routing, Space Network

I. INTRODUCTION

In recent years, large-scale LEO satellite networks have garnered widespread attention as one of the recognized highlights of 6G technology [1]. Modern LEO mega-constellations, consisting of hundreds to thousands of LEO satellites connected by inter-satellite links (ISLs), enable the globally ubiquitous connections [2]. Meanwhile, advancements in remote sensing and chip technology have enabled modern LEO satellites to be equipped with high-performance imaging and intelligent computing payloads [3], [4], thus supporting ubiquitous intelligent services such as weather forecasting, environmental monitoring, and national security [5], [6]. Hundreds to thousands of satellite computing nodes collectively form the SCPN, where the strategic allocation of sensing, communication, and computing resources to efficiently fulfill diverse intelligent service requests has emerged as a critical challenge in the context of 6G integrated space-air-ground networks [7].

Particularly, due to the large-scale deployment of LEO satellites, advancements in satellite remote sensing and data analytics technology, and the high image resolution afforded by the lower orbital altitude, LEO satellite remote sensing

services have demonstrated unprecedented momentum. These services continuously generate massive amounts of data featuring multiple formats and modalities, catering to extensive user demands [8]–[10]. This paper illustrates these application scenarios as depicted in Fig. 1.

Due to the enormous and continuously growing volume of remote sensing data, downloading all satellite data to the ground for centralized processing consumes substantial satellite-to-ground communication resources and may cause congestion for other services. Therefore, leveraging onboard computational resources to perform partial or complete processing of remote sensing tasks and then downloading the computational results to the ground can save communication resources and mitigate privacy and security risks associated with downloading raw data [11]. Furthermore, considering the potential computational resource constraints of a single satellite, the multi-satellite collaborative computing paradigm has emerged as a mainstream research focus [12], [13]. Specifically, thanks to the development of containerization technologies and the evolution of microservice architectures [14]–[16], the functional modules of the user applications, which are traditionally deployed in terrestrial data centers to process raw data postback from satellites, [17], can now be decomposed into stateless microservices and packaged as independent and isolated containers [18]. These services can then be dispatched to computing nodes of the SCPN for on-board computation so as to reduce the transmission delay caused by limited bandwidth and vulnerable channel conditions between ground and space networks [19], [20].

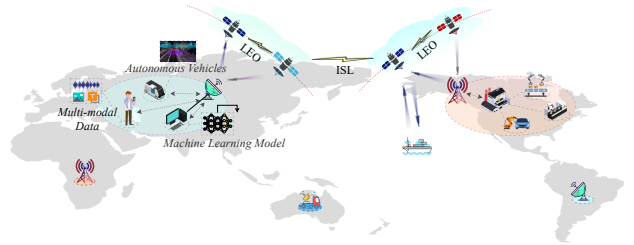


Fig. 1: Space and terrestrial network topology.

However, unlike terrestrial computing nodes with adequate computing resources and power supply. The SCPN nodes present unique limitations, i.e., constrained computing re-

sources and poor power supply. On the one hand, satellites can only be equipped with limited computing and storage resources as a consequence of the constraints of size and weight, as well as the damage risk caused to circuit boards by intense radiation in space [21]. On the other hand, the energy supply for satellites is primarily derived from solar energy harvesting, which is utilized for various satellite functions, including attitude control, transceivers, cameras, and processors. Furthermore, considering the area of solar panels, the efficiency of the materials used for energy harvesting, and hardware degradation, the battery power available for satellites is extremely limited compared to the energy supply available to terrestrial servers. [22], [23].

Hence, different from terrestrial settings [24]–[26], the energy efficiency should be given high priority and regarded as the main concern when devising the routing schemes in the SCPN [22], [27], [28]. To this end, this work presents an energy-aware service routing algorithm for microservices in the SCPN, aiming to minimize energy consumption in both computation and communication to meet the task requirements of terrestrial clients. Specifically, we first model the service routing problem as a Directed Steiner Tree (DST) on the directed graph, which is a classic combinatorial optimization problem and has been proved to be NP-hard [29], [30], then design a series of heuristic-based algorithms to obtain the approximate optimal solution for the service routing problem in the SCPN.

Our main contributions are summarized as follows:

- **Service Routing Modeling in the SCPN:** We identify satellite energy consumption as a critical bottleneck of microservice routing in the SCPN. To this end, we meticulously model the satellite energy consumption associated with microservice communication and computation as well as mathematically formulate the energy resource optimization problem for satellite microservices.
- **Augmented Satellite-Microservice Subgraph Construction:** We propose an innovative augmented satellite-microservice subgraph construction method, which equivalently transforms the original problem into a classical DST problem in graph theory, thereby elucidating the mathematical essence of this engineering challenge. In addition to microservice scenarios, this method is also applicable to other routing problems that jointly optimize the computational and communication resource costs.
- **Heuristic Algorithms for the DST Problem:** We implement a series of heuristic algorithms to approximate the optimal solution for the NP-hard DST problem. Extensive experiments demonstrate that our proposed algorithm not only significantly outperforms baseline algorithms in energy efficiency but also exhibits remarkable stability, thereby validating the efficacy of our augmented graph construction methodology.

The remainder of this paper is organized as follows. Section II reviews the related works. Section III presents the system models. The problem formulation is introduced in Section IV.

Section V details the proposed heuristic algorithms. Section VI shows and analyses the experiment results. Section VII draws the conclusion.

II. RELATED WORK

Recent advancements in satellite technology have significantly promoted its computing and transmitting capabilities, enabling satellites to undertake some elementary tasks [8]. The construction of ISLs, in particular, enables interconnected satellites to form the SCPN with pooled computing resources, supporting intricate missions with enhanced performance demands [13], [21], [31] and facilitating a global response to requests from anywhere at any time [32], [33]. In addition, careful energy management is necessary for satellites to realize sustainable development [22], [23].

Considering the discrete distributed computing nodes (as satellites) with limited computing power in the SCPN, researchers have migrated the monolithic applications (such as 3D reconstruction and inference model), which are traditionally deployed on ground data centers, to space networks by decomposing them into microservices, which are characterized by being independent, scalable, reusable, and stateless [16], [18], [34]. These microservices can be scaled and deployed on the satellites through platform-independent containerization technology, which encapsulates these isolated microservices into containers that are transparent to the installed computing platform on satellites [14], [35].

Extensive research has been conducted on service routing algorithms, aiming at the problems of node selection and routine planning. Fu et al. [28] proposed a dynamic programming approach to drive the optimal policy on energy allocation for a single satellite in response to user requests. It fails to consider the multi-satellite cases. The work [36] designed a joint edge server placement and service deployment model for SAGIN, aiming to minimize the average response latency and energy cost. The work [22] developed a series of heuristic algorithms to prolong the serving life of satellites by switching part of routers installed on satellites to sleep mode, and resolving the energy-efficient routing policies. Efficient but shortsighted, these proposed algorithms cannot adapt to diverse routing needs, for example, the treed topology raised by requested missions, which demands integrated computing and communication routing as well as multi-satellite collaboration.

III. SYSTEM MODEL

We consider a space computing power network (SCPN) composed of a Walker-Delta LEO constellation, onboard computing resources, and laser inter-satellite links (ISLs). User tasks are represented as ordered microservice chains. Each microservice can have multiple stateless replicas deployed on different satellites, and a request may therefore be executed collaboratively by multiple satellites. The key distinction from terrestrial edge systems is that the SCPN simultaneously exhibits predictable but fast topology dynamics, intermittent cross-plane ISLs, limited onboard compute and storage resources, and time-varying communication conditions.

A. LEO Constellation and Time-slotted Topology

We consider a Walker-Delta constellation parameterized by $\mathcal{W} = (A/P/G, H, I)$, where A is the total number of satellites, P is the number of orbital planes, G is the inter-plane phasing factor, H is the orbital altitude, and I is the inclination. Consequently, each orbital plane accommodates $N = A/P$ satellites. A satellite node is uniquely identified by its orbital plane index p and its intra-plane position index n , denoted by $v_{p,n}$. The set of satellite nodes in the constellation is thus formally defined as $\mathcal{V} = \{v_{p,n} \mid p = 1, \dots, P, n = 1, \dots, N\}$. To capture the high dynamism of the satellite network, we discretize the constellation's operational period into a set of time slots, denoted by $\mathcal{T} = \{0, 1, \dots, T-1\}$, with a uniform slot duration Δ . For any continuous-time instant τ , the corresponding time slot index is mapped as $s(\tau) = \lfloor \tau/\Delta \rfloor \bmod T$. Adopting a snapshot-based modeling approach, we assume that the network state, including satellite positions, ISL availability, and nominal link capacities, remains quasi-static within a single time slot, and only evolves at the slot boundaries governed by orbital mechanics. Thus, the time-varying network topology during slot $t \in \mathcal{T}$ is formally defined as a dynamic graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{E}(t)$ represents the set of feasible ISLs at slot t .

B. ISL Communication Model

We assume that each satellite is equipped with four laser optical terminals: two positioned along the roll axis for intra-plane ISLs, and two along the pitch axis for cross-plane ISLs.

a) ISL Connectivity Constraints: Intra-plane ISLs are connected by a satellite $v_{p,n}$ to its adjacent neighbors $v_{p,n-}$ and $v_{p,n+}$ within the same orbital plane, where $n^\pm = ((n \pm 1 - 1) \bmod N) + 1$. Since satellites within the same plane share identical orbital velocities, their relative physical distances remain invariant, forming highly stable ISLs with each other.

Conversely, cross-plane ISLs connecting satellites in adjacent orbital planes are inherently dynamic. Constrained by the physical and kinematic limits of onboard laser communication terminals (LCTs), the availability of a cross-plane link $e = (u, v)$ jointly depends on line-of-sight (LoS) occlusion, mechanical tracking thresholds, and polar exclusions. Formally, its existence at time slot t is defined as:

$$e \in \mathcal{E}(t) \iff \begin{cases} d_{u,v}(t) \leq d_{\max}, \\ \theta_{u,v}(t) \leq \theta_{\max}, \\ \max(|\varphi_u(t)|, |\varphi_v(t)|) \leq \varphi_{\max}. \end{cases} \quad (1)$$

In (1), $d_{u,v}(t)$ represents the LoS distance between satellites u and v at time slot t , and $d_{\max}$ represents the theoretical maximum distance before terrestrial occlusion. The variable $\theta_{u,v}(t)$ denotes the relative tracking angle, which is bounded by the maximum tolerable hardware threshold $\theta_{\max}$. The third constraint enforces a polar exclusion zone parameterized by a latitude boundary $\varphi_{\max}$ (e.g., 70°). This mechanism disconnects satellites entering polar regions, mitigating the severe pointing dynamics and extreme Doppler shifts inherent to high-inclination orbits.

b) ISL Capacity Model: Unlike ground networks with stable capacities, the transmission rate of laser ISLs is distance-dependent due to geometric propagation loss. Under the Free-Space Optical (FSO) communication model, the received optical power $P_e^{\text{rx}}(t)$ at satellite v transmitted from satellite u is formalized as:

$$P_e^{\text{rx}}(t) = P^{\text{tx}} G^{\text{tx}} G^{\text{rx}} \eta_{\text{tx}} \eta_{\text{rx}} \left(\frac{\lambda}{4\pi d_{u,v}(t)} \right)^2, \quad (2)$$

where P^{tx} is the nominal laser transmit power, G^{tx} and G^{rx} represent the optical gains of the transmitter and receiver apertures respectively, λ is the laser wavelength, and $\eta_{\text{tx}}, \eta_{\text{rx}}$ denote the optical efficiency factors. Consequently, the real-time Signal-to-Noise Ratio (SNR) of the link e at time slot t is formulated as:

$$\gamma_e(t) = \frac{(R_{\text{pd}} \cdot P_e^{\text{rx}}(t))^2}{\sigma_{\text{sh}}^2 + \sigma_{\text{th}}^2}, \quad (3)$$

where R_{pd} is the photodiode responsivity, while σ_{sh}^2 and σ_{th}^2 denote the shot and thermal noise variances, respectively. According to the Shannon-Hartley theorem, the achievable transmission capacity $R_e(t)$ is constrained by:

$$R_e(t) = B_e \log_2(1 + \gamma_e(t)), \quad (4)$$

where B_e is the modulation bandwidth.

The nominal rate in (4) is further affected by transient channel fluctuation and traffic contention. We capture this effect through a rate discount factor $\eta_e(t) \in (0, 1]$ and define the effective rate as

$$R_e^{\text{eff}}(t) = \eta_e(t) R_e(t). \quad (5)$$

When a data block of size B is transmitted over link e in slot t , the link-level communication delay consists of waiting delay, transmission delay, and propagation delay:

$$T_e^{\text{comm}}(B, t) = \tau_e^{\text{wait}}(t) + \frac{B}{R_e^{\text{eff}}(t)} + \tau_e^{\text{prop}}(t). \quad (6)$$

Here, $\tau_e^{\text{wait}}(t)$ captures the waiting time caused by the pending transmission queue on link e , and can be estimated from lightweight queue-length probing or historical link-load statistics. The term $B/R_e^{\text{eff}}(t)$ is the transmission delay, which jointly depends on the data volume and the current rate discount factor, while $\tau_e^{\text{prop}}(t) = d_e(t)/c$ is the propagation delay. This compact model allows ELARA to account for link contention without explicitly modeling every concurrent background flow.

C. Onboard Computing Model

Each satellite v has nominal computing capacity F_v . Its real-time service capacity can be lower than F_v because onboard payload tasks, operating-system services, thermal throttling, and other co-running workloads consume part of the processor budget. We capture this effect through a compute discount factor $\xi_v(t) \in (0, 1]$ and define

$$F_v^{\text{eff}}(t) = F_v \xi_v(t). \quad (7)$$

Let $Q_v^{\text{comp}}(t)$ denote the pending compute queue observed on satellite v at slot t . The waiting time before executing a new microservice is denoted by $W_v(t)$, which is an increasing function of $Q_v^{\text{comp}}(t)$. For microservice m with CPU demand C_m , the computation delay on satellite v consists of waiting delay and execution delay:

$$T_m^{\text{comp}}(v, t) = W_v(t) + \frac{C_m}{F_v^{\text{eff}}(t)}, \quad (8)$$

and

$$E_m^{\text{comp}}(v, t) = P_v^{\text{comp}} \frac{C_m}{F_v^{\text{eff}}(t)}, \quad (9)$$

where P_v^{comp} is the active computing power.

D. Microservice Replicas and Service Requests

The application functionality is decomposed into a set of stateless microservices

$$\mathcal{M} = \{m_1, m_2, \dots, m_M\}. \quad (10)$$

Each microservice m has CPU demand C_m , container image size I_m , and storage demand S_m . Let

$$x_{m,v}(t) = \begin{cases} 1, & \text{if a replica of } m \text{ is deployed on } v \text{ in slot } t, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

The replica set of m is

$$\mathcal{R}_m(t) = \{v \in \mathcal{V} \mid x_{m,v}(t) = 1\}. \quad (12)$$

Replica deployment satisfies

$$r_m^{\min} \leq \sum_{v \in \mathcal{V}} x_{m,v}(t) \leq r_m^{\max}, \quad \forall m \in \mathcal{M}, \quad (13)$$

and

$$\sum_{m \in \mathcal{M}} x_{m,v}(t) S_m \leq S_v^{\max}, \quad \forall v \in \mathcal{V}. \quad (14)$$

The stateless assumption allows replica addition, removal, and migration without inter-replica state synchronization.

An incoming request is an ordered service chain

$$r = \langle v_s, m_1, m_2, \dots, m_L, v_d, \tau_{\text{arr}} \rangle, \quad (15)$$

where v_s and v_d are the source and destination satellites, m_i is the i -th required microservice, L is the chain length, and τ_{arr} is the continuous arrival time. Request r has a deadline D_r . Let $B_{r,i}$ be the data volume transmitted before executing m_i , and let $B_{r,L+1}$ be the final output data transmitted from the last selected replica to v_d . The service chain is non-pipelined: m_{i+1} can start only after m_i has completed and the corresponding intermediate data has arrived.

E. Cross-slot Execution Model

ELARA maintains each service chain in continuous time. Before executing m_i , the request state is

$$\sigma_{r,i} = (u_{r,i}, \tau_{r,i}, i), \quad (16)$$

where $u_{r,i}$ is the satellite currently holding the data and $\tau_{r,i}$ is the decision time. The controller selects an execution satellite

$$v_{r,i} \in \mathcal{R}_{m_i}(s(\tau_{r,i})). \quad (17)$$

The data $B_{r,i}$ is routed from $u_{r,i}$ to $v_{r,i}$. Since transmission may cross slot boundaries, routing is recomputed whenever the slot changes. Let $T_{r,i}^{\text{route}}$ and $E_{r,i}^{\text{route}}$ denote the communication delay and energy for this hop. The data arrival time at the selected replica is

$$\tau_{r,i}^{\text{arr}} = \tau_{r,i} + T_{r,i}^{\text{route}}. \quad (18)$$

The computing completion time is

$$\tau_{r,i}^{\text{fin}} = \tau_{r,i}^{\text{arr}} + T_{m_i}^{\text{comp}}(v_{r,i}, s(\tau_{r,i}^{\text{arr}})). \quad (19)$$

Then $u_{r,i+1} = v_{r,i}$ and $\tau_{r,i+1} = \tau_{r,i}^{\text{fin}}$. After m_L finishes, the output data is routed from $v_{r,L}$ to v_d . The end-to-end latency is

$$T_r^{\text{e2e}} = \tau_r^{\text{out}} - \tau_{\text{arr}}, \quad (20)$$

where τ_r^{out} is the time when the final output reaches v_d .

IV. PROBLEM FORMULATION

We formulate energy- and latency-aware service routing with adaptive replica redeployment. The online controller jointly determines which replica executes each microservice, how data is routed across time-varying ISLs, and when service replicas should be added, removed, or moved. The formulation captures the ELARA design without assuming full knowledge of future requests, future queue states, or future link-rate fluctuations.

A. Decision Variables

For request r , stage i , and satellite v , the fast-layer execution decision is

$$z_{r,i,v} = \begin{cases} 1, & \text{if } m_i \text{ of } r \text{ is executed on } v, \\ 0, & \text{otherwise.} \end{cases} \quad (21)$$

Each stage must be executed by exactly one available replica:

$$\sum_{v \in \mathcal{V}} z_{r,i,v} = 1, \quad \forall r, i, \quad (22)$$

and

$$z_{r,i,v} \leq x_{m_i,v}(s(\tau_{r,i})), \quad \forall r, i, v. \quad (23)$$

For routing hop i of request r , let $f_{r,i,e}(t) \geq 0$ be the amount of data transmitted through link e in slot t . Its capacity constraint is

$$0 \leq f_{r,i,e}(t) \leq R_e^{\text{eff}}(t) \Delta_{r,i}^{\text{rem}}(t), \quad \forall e \in \mathcal{E}(t), \quad (24)$$

where $\Delta_{r,i}^{\text{rem}}(t)$ is the remaining usable time in slot t . At each slot boundary, the residual data is updated and the routing decision is recomputed.

The slow-layer redeployment action in slot t is

$$a_t \in \{\text{add, remove, move, no-op}\}. \quad (25)$$

Every action must preserve (13) and (14). For add and move actions, the target satellite must have enough residual storage and must not already host the same microservice. For remove actions, the service must still have at least $r_m^{\min}$ replicas after removal.

B. Latency, Energy, and Migration Cost

For one unit of data on link e in slot t , the unit delay cost follows the communication-delay decomposition in (6):

$$c_e^T(t) = \frac{1}{R_e^{\text{eff}}(t)} + \tau_e^{\text{prop}}(t) + \tau_e^{\text{wait}}(t) + \tau^{\text{sw}}, \quad (26)$$

where τ^{sw} denotes per-hop switching overhead. The unit communication energy is

$$c_e^E(t) = P_e^{\text{tx}} \frac{1}{R_e^{\text{eff}}(t)}. \quad (27)$$

Let $\phi_e(t)$ be the short-horizon link disruption risk estimated from future ISL availability over a small look-ahead window. The routing-layer edge cost is

$$c_e(t) = \omega_T c_e^T(t) + \omega_E c_e^E(t) + \omega_R \phi_e(t). \quad (28)$$

For request r , the total communication delay and energy are

$$T_r^{\text{route}} = \sum_{i=1}^{L+1} T_{r,i}^{\text{route}}, \quad E_r^{\text{route}} = \sum_{i=1}^{L+1} E_{r,i}^{\text{route}}. \quad (29)$$

The computation delay and energy are

$$T_r^{\text{comp}} = \sum_{i=1}^L \sum_{v \in \mathcal{V}} z_{r,i,v} T_{m_i}^{\text{comp}}(v, s(\tau_{r,i}^{\text{arr}})), \quad (30)$$

and

$$E_r^{\text{comp}} = \sum_{i=1}^L \sum_{v \in \mathcal{V}} z_{r,i,v} E_{m_i}^{\text{comp}}(v, s(\tau_{r,i}^{\text{arr}})). \quad (31)$$

Thus,

$$E_r^{\text{total}} = E_r^{\text{route}} + E_r^{\text{comp}}. \quad (32)$$

For a replica redeployment action a_t , the migration cost is

$$C^{\text{mig}}(a_t) = \omega_T^{\text{mig}} \frac{T^{\text{mig}}(a_t)}{T_0} + \omega_E^{\text{mig}} \frac{E^{\text{mig}}(a_t)}{E_0}, \quad (33)$$

where $T^{\text{mig}}(a_t)$ and $E^{\text{mig}}(a_t)$ are the delay and energy of transferring the service image. Remove and no-op actions have zero migration-transfer cost.

C. Optimization Objective

Let N_r^{cross} be the number of slot crossings experienced by request r , and let $\mathbb{I}_r^{\text{fail}}$ indicate whether the request fails. The per-request cost is

$$J_r = \omega_T \frac{T_r^{\text{e2e}}}{T_0} + \omega_E \frac{E_r^{\text{total}}}{E_0} + \omega_S N_r^{\text{cross}} + \omega_F \mathbb{I}_r^{\text{fail}}. \quad (34)$$

Given a sequence of requests $\mathcal{Q}$, the online problem is

$$\mathbf{P1} : \min_{\mathbf{z}, \mathbf{f}, \mathbf{a}} \sum_{r \in \mathcal{Q}} J_r + \omega_M \sum_{t \in \mathcal{T}} C^{\text{mig}}(a_t) \quad (35a)$$

$$\text{s.t.} \quad (22), (23), \quad (35b)$$

$$0 \leq f_{r,i,e}(t) \leq R_e^{\text{eff}}(t) \Delta_{r,i}^{\text{rem}}(t), \quad \forall r, i, e, t, \quad (35c)$$

$$\sum_{e \in \delta^+(u)} f_{r,i,e}(t) - \sum_{e \in \delta^-(u)} f_{r,i,e}(t) = b_{r,i,u}(t), \quad \forall r, i, u, t, \quad (35d)$$

$$(13), (14), \quad (35e)$$

$$a_t \in \{\text{add, remove, move, no-op}\}, \quad \forall t, \quad (35f)$$

$$T_r^{\text{e2e}} \leq D_r \quad \text{or} \quad \mathbb{I}_r^{\text{fail}} = 1, \quad \forall r. \quad (35g)$$

In (35d), $b_{r,i,u}(t)$ is the net flow balance for routing hop i of request r : it is positive at the current source, negative at the selected execution or destination node, and zero at relay satellites. The formulation is online because the controller observes the current topology, deployment state, link queues, and compute queues, while future requests and future queue dynamics are unknown.

D. Problem Hardness

Proposition 1. Problem **P1** is NP-hard.

Sketch. Consider a restricted case with a single time slot, fixed replica deployment, fixed effective link rates and compute capacities, and no redeployment. The controller still needs to select one replica for each service in a chain and connect consecutive selected replicas with feasible paths to minimize joint computation and communication cost. This restricted case contains the classical service placement and routing problem as a special case and can encode constrained shortest-path and facility-location-like decisions. Therefore, the restricted problem is NP-hard, and the full problem with cross-slot routing, stochastic load, and adaptive redeployment is at least as hard.

Solving **P1** exactly is not suitable for online SCPN operation because requests arrive continuously, topology changes every slot, and redeployment decisions have delayed effects. ELARA therefore decomposes the problem into three coupled online layers: a fast PPO-GNN replica selector, a cross-slot min-cost max-flow router, and a slow bandit-based replica redeployment agent.

V. ENERGY-AWARE SERVICE ROUTING ALGORITHM FOR MICROSERVICES IN THE SCPN

In this section, we present our approach to addressing the proposed problem **P** as outlined in Section IV. We first detail our augmented satellite-microservice connected sub-graph

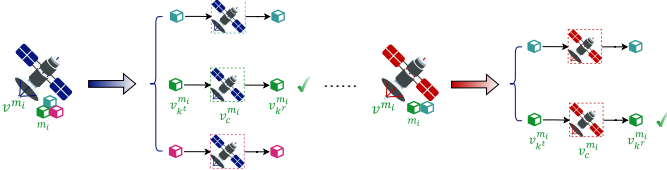


Fig. 2: Abstract the computation process from colocated microservice and satellite deployment.

Algorithm 1: Augmented Satellite-Microservice Connected Subgraph Construction Algorithm.

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1 Initialize augmented directed graph  $\mathcal{G}_z$  with vertex set
 $\mathcal{V}_z \leftarrow \emptyset$  and edge set  $\mathcal{E}_z \leftarrow \emptyset$ ;
2  $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \{v^{src}, v^{dst_1}, \dots, v^{dst_\rho}\}$ ;
3 for  $m_i$  in  $\mathcal{M}_f$  do
4   for  $v_k$  in  $\mathcal{V}_{m_i}$  do
5      $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \{v_{k^r}^{m_i}, v_{k^c}^{m_i}, v_{k^t}^{m_i}\}$ ;
6      $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \{(v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}), (v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i})\}$ ;
7     Set  $w(v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}) \doteq E_{m_i}(v_k)/2$ ;
8     Set  $w(v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i}) \doteq E_{m_i}(v_k)/2$ ;
9   end
10 end
11 for  $w_{i,j}$  in  $\mathcal{W}_f$  do
12   for  $v_k^{m_i}$  in  $\mathcal{V}_{m_i}$  do
13     for  $v_k^{m_j}$  in  $\mathcal{V}_{m_j}$  do
14        $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v_k^{m_i}, v_k^{m_j})$ ;
15        $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
16        $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
17     end
18   end
19 end
20 for  $v_k^{m_{src}}$  in  $\mathcal{V}_{m_{src}}$  do
21    $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v^{src}, v_k^{m_{src}})$ ;
22    $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
23    $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
24 end
25 for  $v_k^{m_{dst}}$  in  $\mathcal{V}_{m_{dst}}$  do
26   for  $g$  in  $[1, \rho]$  do
27      $\mathcal{V}_o^{re}, \mathcal{E}_o^{re} = \text{GISRSA}(\mathcal{G}, v_k^{m_{dst}}, v^{dst_g})$ ;
28      $\mathcal{V}_z \leftarrow \mathcal{V}_z \cup \mathcal{V}_o^{re}$ ;
29      $\mathcal{E}_z \leftarrow \mathcal{E}_z \cup \mathcal{E}_o^{re}$ ;
30   end
31 end

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construction algorithm, and then elaborate on the design of our DST-based service routing algorithm tailored to optimize energy consumption in the SCPN.

A. Augmented Satellite-Microservice Sub-Graph Construction

For each request originating from a terrestrial station, there is a DAG $\mathcal{G}_f^m$, which outlines the service execution sequence. In $\mathcal{G}_f^m$, there may exist ρ ($1 \leq \rho \leq |\mathcal{M}_f| - 1$) vertices whose outdegree equals 0. These vertices represent terminal microservices that conclude the processing sequence by dispatching

results to their respective destinations. Consequently, there should be multiple logical destination satellite nodes, denoted as $\{v^{dst_1}, \dots, v^{dst_\rho}\}$. The routing paths from the logical destination satellites hosting the terminal microservices to the real destination satellites can be straightforwardly generated by the control panel. Note that if a terminal microservice is deployed directly on its real destination satellite, this satellite also acts as the logical destination. In such cases, the communication cost between the terminal service and its destination is effectively zero. Next, we define the augmented graph to be constructed as $\mathcal{G}_z = (\mathcal{V}_z, \mathcal{E}_z)$, where initially, $\mathcal{V}_z = \emptyset$ and $\mathcal{E}_z = \emptyset$. Then, the process of constructing graph $\mathcal{G}_z$ is detailed as follows:

Step 1. This step involves adding the source node v^{src} , along with the logical destination nodes $\{v^{dst_1}, \dots, v^{dst_\rho}\}$ and the real destination node v_{dst} , to vertex collection $\mathcal{V}_z$.

Step 2. For each microservice m_i in $\mathcal{M}_f$, and $\forall v_k \in \mathcal{V}_{m_i}$, this step creates three distinct vertices: $v_{k^r}^{m_i}$, $v_{k^c}^{m_i}$ and $v_{k^t}^{m_i}$, which are then added to the set $\mathcal{V}_z$. Subsequently, it constructs two directed edges $v_{k^r}^{m_i} \rightarrow v_{k^c}^{m_i}$ and $v_{k^c}^{m_i} \rightarrow v_{k^t}^{m_i}$, both with weights equal to $E_{m_i}(v_k)/2$. These edges are added to the edge set $\mathcal{E}_z$. This procedure is detailed in Fig. 2.

Step 3. For the calling relationship between two microservices m_i and m_j , that is, $\forall w_{i,j} \in \mathcal{W}_f$, the algorithm will iteratively identify all satellite nodes from collection $\mathcal{V}$ that host microservices m_i and m_j , respectively. If satellites $v_k^{m_i}$ and $v_k^{m_j}$ are directly adjacent without needing any relay satellite to facilitate the communication, then a directed edge $v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j}$ is constructed, and its weight is set as the communication energy cost from $v_{k^r}^{m_i}$ to $v_{k^t}^{m_j}$ in $\mathcal{G}$, given as:

$$e^{tr} = P_T^{v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j}} \frac{w_{i,j}}{w(v_{k^r}^{m_i} \rightarrow v_{k^t}^{m_j})}, \quad (36)$$

Otherwise, the algorithm then searches for available relay satellite nodes within $\mathcal{V}$, which can be efficiently resolved using the Floyd algorithm, resulting in a collection of relay satellite nodes denoted as $\mathcal{V}^{re} = \{v_1^{re}, \dots, v_\xi^{re}\}$. Then, it adds these relay nodes to $\mathcal{V}_z$, and constructs and adds edges $v_{k^t}^{m_i} \rightarrow v_a^{re}, v_a^{re} \rightarrow v_{a+1}^{re}, v_\xi^{re} \rightarrow v_{k^r}^{m_j}, (\forall a \in [1, \xi - 1])$ to $\mathcal{E}_z$. The weights of these edges, representing the relay communication energy cost between satellites, are calculated according to Eq. ?? and ??, as below:

$$e_{re1}^{tr}(v_{k^t}^{m_i} \rightarrow v_a^{re}) = P_T^{v_{k^t}^{m_i} \rightarrow v_a^{re}} \frac{w_{i,j}}{w(v_{k^t}^{m_i} \rightarrow v_a^{re})}, \quad (37)$$

$$e_{re2}^{tr}(v_a^{re} \rightarrow v_{a+1}^{re}) = P_T^{v_a^{re} \rightarrow v_{a+1}^{re}} \frac{w_{i,j}}{w(v_a^{re} \rightarrow v_{a+1}^{re})}, \quad (38)$$

$$e_{re3}^{tr}(v_\xi^{re} \rightarrow v_{k^r}^{m_j}) = P_T^{v_\xi^{re} \rightarrow v_{k^r}^{m_j}} \frac{w_{i,j}}{w(v_\xi^{re} \rightarrow v_{k^r}^{m_j})}. \quad (39)$$

This process is depicted in Fig. 3.

Step 4. For the first executed microservice m_{src} and the final executed microservices $\{m_{dst_1}, \dots, m_{dst_\rho}\}$, the algorithm searches for the collection of satellite nodes hosting these microservices. Subsequently, it constructs the edges $v^{src} \rightarrow v_{k^r}^{m_{src}}$ and $v_{k^t}^{m_{dst}} \rightarrow v^{dst_g}, \forall g \in [1, \rho]$. The detailed procedures for the construction of the newly generated vertices and edges are similar to those outlined in Step 3.

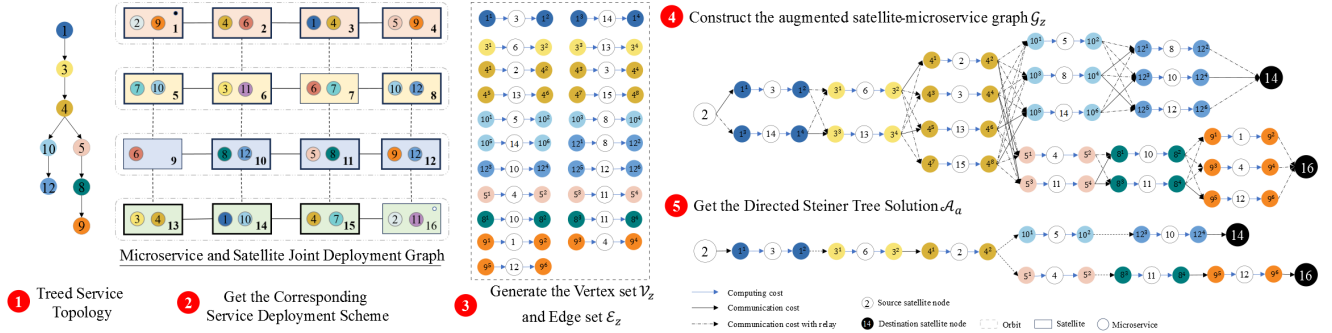


Fig. 4: The overview of solution workflow of tree-based request topology.

minimum spanning Steiner arborescence for the snapshot s , denoted by $\mathcal{A}(s) = (\mathcal{V}_a(s), \mathcal{E}_a(s))$. This step provides an approximate solution to the DST problem, optimizing the cost while ensuring connectivity among specified nodes.

Algorithm 2 presents a detailed step-by-step procedure for the proposed approach, offering a comprehensive view of its implementation. To enhance understanding, a simplified example is provided in Fig. 4, which visually demonstrates the key steps and logic of the algorithm.

C. Algorithm Complexity Analysis

The complexity of Algorithm 1 and 2 is analyzed as follows.

Complexity of Algorithm 1. The time complexity of Dijkstra algorithm is given by $\mathcal{O}(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|)$. In the worst case, the algorithm needs to run $|\mathcal{V}^*|$ times, which is the number of destination satellites, and the total time complexity deteriorates to $\mathcal{O}(|\mathcal{V}^*|(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|))$. This signifies that the complexity is contingent upon the structure of both the service request graph and the space network topology.

Complexity of Algorithm 2. The time complexity of merging these distinct sets into one set is $\mathcal{O}(n)$, where n is the total number of elements across all sets. Therefore, the time complexity of obtaining the minimum DST is inferred as $\mathcal{O}(|\mathcal{V}^*|(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|) + 2*n + (|\mathcal{V}_z| + |\mathcal{E}_z|))$, where $\mathcal{O}(2*n)$ represents the time complexity of merging the generated vertex and edge sets, and $\mathcal{O}(|\mathcal{V}_z| + |\mathcal{E}_z|)$ is the time complexity of constructing the directed Steiner tree.

VI. PERFORMANCE EVALUATION

A. Experiment Setup

Constellation. In experiments, we consider a 72/12/1 Telesat Star Constellation system, which consists of $P = 6$ polar orbit planes whose inclination is 99.5° , and the orbital altitude is $h = 1000km$ on the ground [37].

Satellite Computing Capability. We suppose LEO satellites are equipped with several onboard processors and storage components. According to [27], we assign the satellite computing capability following a uniform distribution, as $\varphi(v_k) \sim \text{DiscreteUniform}(\{100, 160, 220, 280, 340, 400\})$, while the processor power $\Psi(v_k) = \varphi(v_k) \times 0.01$ in Watts.

ISL Parameters. Similar to settings in [32], [33], [38], the transmission power of all relevant satellites is standardized at $P_T = 3$ (Watt). In the context of the Telesat Constellation,

which achieve inter-satellite interconnection across free-space optical link, the carrier frequency of each of satellite is set as $f_c = 193(THz)$, and the corresponding carrier wavelength is $\lambda = 1550(nm)$ while the channel bandwidth between two satellites is defined as $BW = 2\%f_c$. Besides, we specify the antenna diameter transceiver as $D_R = 6(mm)$, the pointing loss angle $\theta_0 = 0.01\pi$ and the optical conversion efficiency of the transceiver modules $\eta = 0.8$.

Microservices. According to the microservice setting adopted by [27], we heuristically construct the microservices testbed as follows: There are $|\mathcal{M}| = 60$ microservices whose data volume follows a Gaussian distribution with a mean of 50 and variance of 4 in Gbits. For each microservice, there are 4 ~ 6 replicas distributed on satellites, while each satellite can concurrently host at most $\lfloor |\mathcal{M}| \times 6/72 \rfloor$ microservices.

Ablation Solutions: We design the following benchmarks to evaluate the performance of our proposed algorithm.

- 1) **Random:** It randomly selects a satellite for each microservice and uses the Dijkstra algorithm to get the shortest path when dealing with relay links.
- 2) **Computation greedy (Comp-greedy):** It chooses the satellite that offers the minimum computation energy cost for executing each microservice. The relay routing policy is the same as **Random** policy.
- 3) **Communication greedy (Comm-greedy):** It picks the

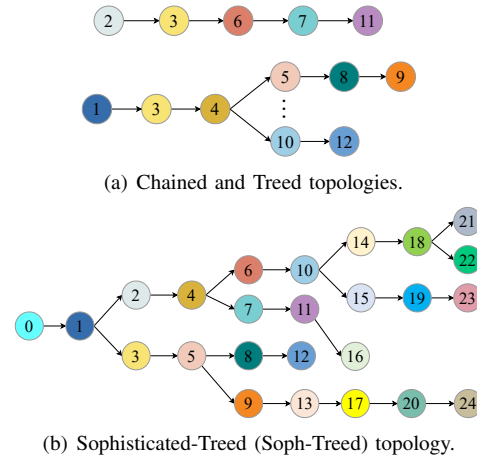


Fig. 5: The overview of service request topologies from terrestrial stations.

satellite pair with the minimum communication energy cost, including the potential relay links.

We define three service request topologies: the Chained topology, the multi-forked Treed topology, and the Sophisticated Treed (Soph Tree) topology, as depicted in Fig. 5.

To verify the effectiveness of our algorithm, we conducted a series of experiments on a space topology comprising 72 satellites, each hosting 60 microservices with 4~6 replicas. We generated 100 service request chains, varying in length from 10 to 20, to simulate a chained microservice topology. We also constructed 100 random treed topologies, containing 20 to 45, featuring a stochastic number of leaf nodes and varying branch configurations. For each sophisticated treed topology, microservices were randomly assigned to nodes, and this assignment process was independently repeated 100 times. For each topology, energy cost data was collected from 100 samples. The final results were derived by averaging the energy cost values across these samples. It ensures robust and reliable performance evaluation by mitigating variations caused by randomness in the topology generation and microservice allocation processes.

B. Experimental Results

1) *Effectiveness*: As illustrated in Fig. 6(a), our proposed Steiner consistently outperforms all baseline methods across various microservices topology scenarios, and its advantages become more pronounced in more complex topologies. This enhanced performance is charged to our comprehensive approach that simultaneously considers both computation and communication costs when determining the most efficient routing paths. The comp-greedy method outperforms the random method by prioritizing minimizing computation costs. The comm-greedy method surpasses both comp-greedy and random methods by focusing on minimizing communication costs. These findings emphasize the necessity of accounting for both computation and communication factors to design effective and energy-conscious routing strategies for the SCPN.

2) *Impact of constellation scales*: To evaluate the scalability of our algorithm, we systematically varied the number of satellites in the constellation by adjusting the number of orbits while maintaining consistent microservice deployment strategies. Experiments were conducted in two constellation settings separately, as illustrated in Fig. 6(b) and Fig. 6(c). These configurations allowed us to assess the algorithm's performance under varying levels of network complexity and service topology. Our results demonstrate that the proposed approach consistently outperforms alternative methods across all tested constellation scales and topological scenarios. Notably, its effectiveness becomes increasingly pronounced as the network complexity escalates, highlighting its robustness and adaptability to intricate environments.

3) *Impact of transmitter power*: Given the dominant role of communication costs in the total energy consumption of satellite systems, we conducted a series of experiments to explore how varying channel conditions influence the performance of our Steiner scheme. Specifically, we experimented

with four different transmitter power levels on satellites across diverse microservice request schemes to examine their effect on energy efficiency. The results shown in Figs. 7(a), 7(b) and 7(c) indicate that as transmitter power increases, the average energy cost shows an upward trend. This trend highlights the trade-off between signal strength and energy consumption, with higher power levels leading to greater energy expenditure. Despite this increase in energy cost, our Steiner algorithm consistently outperformed other benchmark strategies across all service request scenarios. This superior performance is attributed to the algorithm's ability to dynamically adapt to the complexities of space network topologies and fluctuating channel conditions. By optimizing service routing paths, our approach minimizes unnecessary energy use under varying channel conditions. These findings underscore the importance of intelligent service routing mechanisms in the SCPN, particularly in environments with variable communication links.

4) *Impact of faulty satellites*: Given the constraints of limited power supply and the effects of solar eclipses, some satellites may need to enter a low-power mode, functioning solely as relay nodes for message forwarding [22]. To evaluate the robustness of our proposed method under such conditions, we conducted experiments by randomly disabling a subset of satellites in a constellation of 72 satellites. Specifically, we tested scenarios with 10, 20, and 25 faulty satellites by eliminating their microservices, as shown in Figs 8(a), 8(b), 8(c). The results reveal that energy cost generally increases with an increasing number of faulty satellites for all methods. This trend can be attributed to the reduced availability of active nodes, which forces longer transmission paths and higher energy consumption for maintaining connectivity (much more relay satellites are required for message forwarding). Despite this challenge, Steiner invariably outperforms other methods in all scenarios. Its superior performance is attributed to its robust design, which can effectively adapt to changes in network topology and operational conditions. By optimizing power allocation and service routing decisions, the algorithm minimizes disruptions caused by satellite failures.

5) *Impact of microservice deployment schemes*: To investigate the influence of microservice deployment and scaling strategies on energy consumption within the SCPN, we conducted experiments by varying the number of replicas allocated for each microservice. Specifically, we tested three replica ranges: [2, 6], [4, 6], and [6, 10]. And the experimental results, illustrated in Figs. 9(a), 9(b), and 9(c), reveal that energy costs generally decrease as the scale of microservice deployment expands. This trend is attributed to the increased availability of service nodes and routing paths, which provide more opportunities for route optimization and efficient power utilization. Our algorithm regularly demonstrates superior performance in reducing energy costs compared to other benchmarks, particularly in more complex microservice-satellite patterns. This superior performance can be attributed to the algorithm's ability to leverage the increased number of service routing paths and nodes, enabling more advanced route strategies. The expanded deployment range provides greater

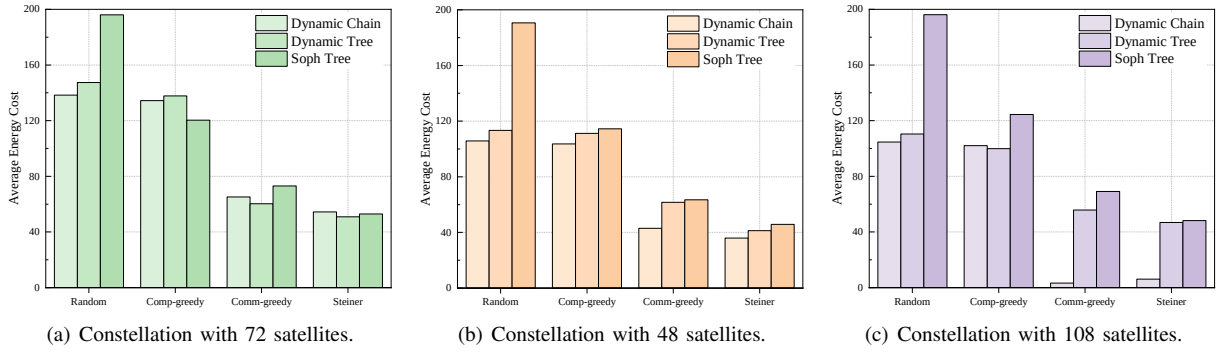


Fig. 6: Performance comparison on energy-cost under varying constellation scales for dynamic microservice request topologies.

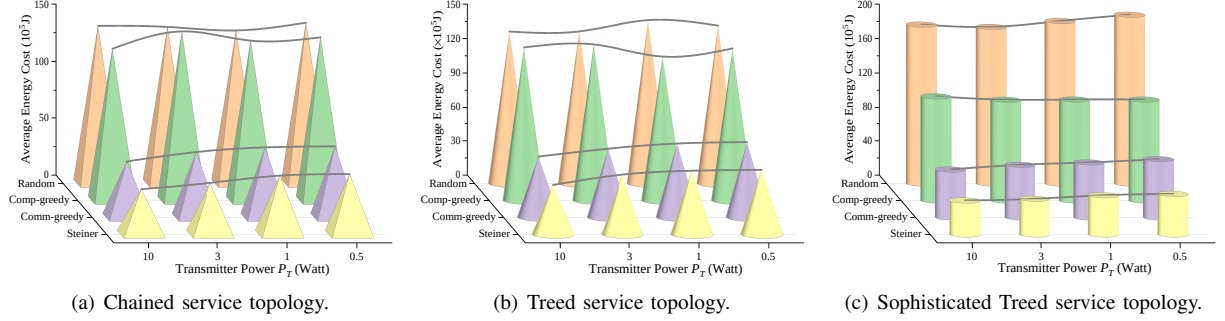


Fig. 7: Performance comparison on energy-cost under different onboard transmitter power configurations for dynamic microservice request topologies.

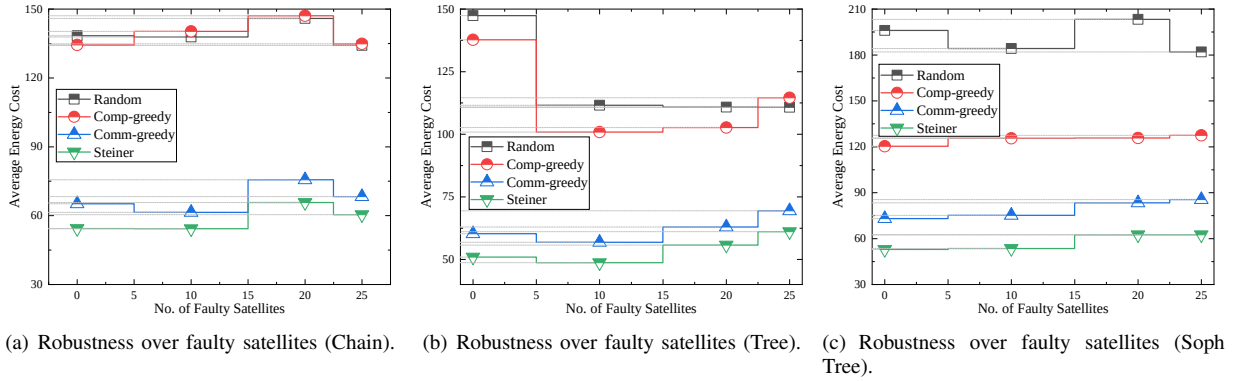


Fig. 8: Performance comparison on energy-cost under different configurations on number of faulty satellites for dynamic service request topologies.

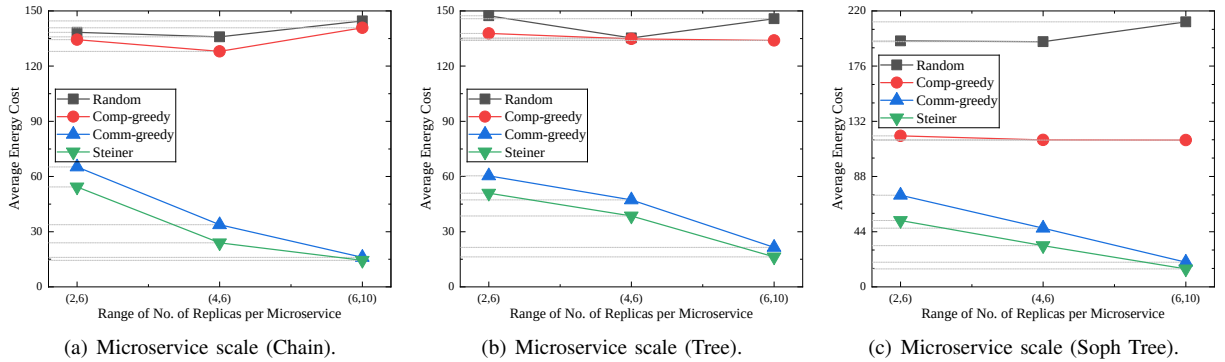


Fig. 9: Performance comparison on energy-cost under different configurations on number of replicas of microservices for dynamic service request topologies.

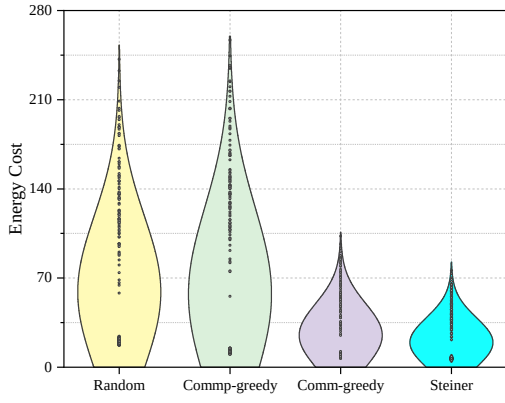


Fig. 10: Energy distribution of Soph Tree.

flexibility in service selections and routing decisions, allowing the algorithm to identify more energy-efficient configurations. By contrast, the energy costs associated with routing paths generated by random and comp-greedy methods occasionally increased as the number of replicas grew. This counterintuitive behavior can be attributed to inefficient routing decisions caused by the higher number of service nodes and calling paths, which complicate the decision-making process and lead to suboptimal resource utilization.

6) *Adaptability and stability*: Furthermore, as depicted in Fig 10, which presents the energy distribution derived from 300 independent experiments, underscores the superior stability of the Steiner policy compared to other benchmark strategies. The results reveal that, even under sophisticated treed topologies with heightened complexity, the Steiner policy exhibits less variance in energy costs. This stability highlights its robustness and adaptability in managing diverse network conditions and service requirements.

VII. CONCLUSION

In this paper, we proposed a DST-based solution to provide an energy-efficient service routing scheme for tasks from terrestrial clients. We systematically developed the SCPN models, including communication and computation, covering various aspects of the LEO satellite network. We also mathematically formulated the energy optimization problem for satellite-microservice routing and transformed it into finding a feasible DST solution within an augmented directed satellite-microservice colocated graph. To approximate the optimal solution, we designed a series of heuristic algorithms. Experimental results show that our algorithm significantly outperforms other benchmarks in energy efficiency, robustness, and adaptability across different network and microservice configurations in the context of SCPN.

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